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Research

Continental-scale shifts in termite diversity and nesting and feeding strategies

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Typically, termites are treated as a single guild, which ignores important internal diversity, including diverse feeding and nesting traits. These termite traits are crucial for both ecosystem-level fluxes and trophic webs, with implications for vertebrate species. Despite their ecological importance, the large-scale distribution of termite feeding and nesting traits and the relationship with termite diversity is largely unknown. We investigated whether functional diversity, species richness and feeding (wood, litter, grass and dung) and nesting trait (aboveground mound, belowground nest, inside tree or outside tree nest) distributions of termites were climatically controlled. To address this gap, we assembled a continental-scale database of termite traits and occurrence in Australia and modelled termite nesting and feeding traits in response to macroclimate. Functional richness and evenness increased primarily with temperature. Australia showed multiple hotspots of termite diversity with each hotspot showing a distinct guild composition. The large-scale distribution of nesting traits showed that aboveground nesting species were the most common nesting guild in the dry and wet tropics, while belowground nesting dominated in seasonally cold arid environments, demonstrating a strong climatic control on nesting strategy. Given their large biomass and many interactions with other species, the macroecology of termite traits may be especially important in predicting shifts in other species' distributions at continental and global scales.

Keywords: Australia, biogeography, climate, functional role, macroecology, termites, traits

Introduction

Termites are eusocial insects belonging to the *Blattodea* order that form an integral part of all but the coldest ecosystems worldwide from temperate forests to tropical savannahs (Wood and Sand 1978, Whitford et al. 1992, Sileshi et al. 2010). Together with a complex array of symbionts inside and (in some cases) outside their guts, termites provide essential ecosystem services as soil ecosystem engineers (Parker et al. 1982, Jouquet et al. 2011) and as both herbivores of highly recalcitrant vegetation



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and decomposers of low-quality litter throughout low- to mid-latitude ecosystems. Current evidence suggests that termites achieve an especially high abundance in savannah and rainforest (Schuurman 2005, Ulyshen 2016, Griffiths et al. 2019). Termites shape ecosystems in many ways such as determining the heterogeneity of vegetation in the landscape (Sileshi et al. 2010, Davies et al. 2016a), mitigating the effects of drought (Ashton et al. 2019) and acting as a food resource for many vertebrates (Kok and Hewitt 1990, Abensperg-Traun 1994). While they are often lumped into one category by ecosystem ecologists, termites are a diverse group of organisms and have a variety of feeding and nesting habits. These functional distinctions among termites have important implications for both nutrient cycling and trophic webs.

To most casual observers, termites are pests feeding on construction timber (Su and Scheffrahn 2000) or the creatures who form fields of mounds in tropical savannas (Spain et al. 1986, Davies et al. 2016a, Martin et al. 2018). In reality, termites are widespread and have a wide range of feeding and nesting habits which overlap throughout diverse habitats. Feeding substrates include wood, grass (*Poaceae* family), litter, dung and soil with some African and Asian species even cultivating fungus as a food source (Eggleton and Tayasu 2001). As eusocial insects, the nest is an important part of termite biology where many important within colony interactions occur (Noirot and Darlington 2000), but there is significant heterogeneity in the types of nest. Some wood-feeding termite species nest within their feeding resource; all others including many wood-feeding species build separate nests (Eggleton and Tayasu 2001), including belowground nests, aboveground epigeal nests usually constructed from clay (hereafter referred to as aboveground mounds) and aboveground arboreal nests (Noirot and Darlington 2000). The location of the nest has important implications for both the spatial distribution of nutrients in the ecosystem and for vertebrate species that prey on termites. For instance, aardvarks *Orycteropus afer* feed on termites in aboveground mounds (Taylor et al. 2002), while slider lizards *Lerista* spp. feed on termites in belowground nests (Greenville and Dickman 2005). Each species of termite will possess a set of feeding and nesting behaviours integrating them in a specific part of the ecosystem.

The feeding and nesting behaviours of termites can be characterized as separate functional traits which define their role in the ecosystem and their interactions with other species (McGill et al. 2006). For instance, aboveground mounds positively influence the availability and quality of vegetation for herbivores across different landscapes (Levick et al. 2010, Davies et al. 2016b), while arboreal nests provide habitats for kingfishers (Hindwood 1959, Kesler and Haig 2005). Wood-feeding termites play a crucial part in the decomposition of wood in forest and savannah ecosystems, competing with fungi for the wood resource (Cornwell et al. 2009, Ulyshen 2016), and grass-feeding termites play a similar role in savannas (Bodine and Ueckert 1975, Silva et al. 1985, Holt and Coventry 1990).

While an increasing amount is known of the role of termites as a group within particular ecosystems (Jouquet et al. 2011), little is known of geographic and climatic distributions of termite traits (Jones and Eggleton 2011). As with work on the functional biogeography of plants first hypothesized and described in von Humboldt's Tableau in the 19th century (von Humboldt and Bonpland 1807) and now for many other taxa including ants, birds and mammals, the distribution of termite traits across space will help build a more complete understanding of what drove and maintained the evolution of termite diversity and their interactions with other species across a wide array of ecosystems.

Previous studies have described the distribution of termite species richness and their functional feeding groups on a global scale (Eggleton 2000, Eggleton and Tayasu 2001, Jones and Eggleton 2011). Taking the global distribution of fungus growers as an example, this feeding trait is exclusively found around the African and East Asian tropics (Eggleton and Tayasu 2001). These species and the efficiency of the fungus-growing strategy plays an integral role in the decomposition of vegetation in African and East Asian ecosystems (Schuurman 2005). While information is available on the distribution of fungus growers on a global scale, less is known of the regional macro-climatic factors determining their distribution and that of other feeding and nesting traits. Several studies have attempted to explain the distribution of specific termite traits but were either focused on a few species (Maynard et al. 2015, Lee et al. 2017, Beasley-Hall et al. 2018) or carried out over 40 years ago when knowledge on termite distribution and traits was sparse (Calaby and Gay 1959, Sen-Sarma 1974). We use the Australian continent as a study system to understand termite macroecology because of its combination of data resources and diverse termite assemblages as well as a broad range of climatic conditions (Sturman and Tapper 1996).

It has long been recognized that termites are a key ecosystem player throughout Australia (Calaby and Gay 1959, Holt and Coventry 1990, Whitford et al. 1992). From the 1940s to the 1990s, the Commonwealth Scientific and Industrial Research Organisation (CSIRO) conducted extensive termite collection campaigns to understand the taxonomy and distribution of termites across Australia (Watson and Abbey 1993). Australian termites include over 250 species (Watson and Gay 1991) with a diverse range of feeding and nesting traits, with a wealth of data on distributions and traits.

The aim of this study is to build a biogeographic understanding of the functional traits of termites, connecting macroclimate to the frequency of different termite nesting and feeding types. This is necessary to build both an understanding of termite functional ecology and an understanding of the different roles of termites across macroclimate gradients. To investigate further, we have decided to focus our study on three response variables: 1) relative probability of occurrence of different feeding and nesting guilds; 2) species richness of termites; and 3) functional diversity. Functional diversity can further be split into functional richness, evenness and divergence. The variables

were assessed against continental abiotic gradients including mean temperatures varying from 3.4°C to 29°C, mean annual precipitations ranging from 148 mm to 4200 mm and other variables including soil clay content and net primary productivity.

Because of the paucity of previous work on the global macroecology of termites (Eggleton 2000, Davies et al. 2003) and their unusually high local diversity in some drier habitats of Australia (Clement et al. 2021), it is not possible to formulate strong hypotheses on termite diversity patterns. As demonstrated previously in the literature, we expect species richness and functional richness to be correlated (Tilman et al. 1997, Bihn et al. 2010). Furthermore, evidence from regional-scale studies suggests that precipitation determines the distribution of termite mounds (Davies et al. 2014) and that wood-feeding species are restricted to habitats with the highest wood resource availability (Maynard et al. 2015). Because the density of woody plants increases with precipitation, one might expect that the frequency of wood-consuming species might also increase with precipitation. Additionally, due to the ectothermic nature of termites, the temperature of their surrounding environment is crucial for their survival and can be manipulated by the structure of their nest for instance (Noirot and Darlington 2000, Korb 2003). As has been observed in ants (Reymond et al. 2013), we expect that subterranean nests in termites should be favoured at lower temperatures to avoid excessive heat loss. Finally, soil properties have also been shown to be an important determinant of termite mound establishment as these structures depend on fine clay soils for their stability (Jouquet et al. 2002, 2004); therefore, we expect that mound-building species should be favoured in high clay content areas.

Methods

Termite distribution data

A total of 17 457 termite specimen occurrence records, from 5 families – *Termitidae*, *Rhinotermitidae*, *Kalotermitidae*, *Stolotermitidae* and *Mastotermitidae* – were collected between 1893 and 2019 across Australia (Fig. 1c). The records were obtained from the Global Biodiversity Information Facility database (GBIF 2019) and represent all the records of termites identified to species for which spatial data were available. Unlike for other taxa where most data are recent, in this data set, most records were collected between the 1940s and the 1990s with a median date of data collected in 1966 (Supporting information). Of the data, 88.9% were collected by the CSIRO (Hill 1942, Calaby 1956, Watson and Abbey 1993), 9.4% by various museum collections across Australia, 0.7% from government environmental agencies, 0.1% from citizen science and 0.9% from unknown sources (these were subsequently removed to avoid using unreliable species records). We assume that every record is different as we could not control for potential mismatches or naming mistakes in the database. The temporal pattern in the data set arises from an extensive and ambitious termite collection campaigns by CSIRO using a combination of horses and cars to traverse Australia actively and opportunistically looking for any signs of termites (Fig. 1a–b) along what constituted major roads or trails at that time. As with almost all biodiversity data sets, more detailed termite collection campaigns with thorough methodologies were undertaken closer to inhabited areas and along roads compared to more remote locations (Watson 1988, Watson et al. 1989, Abensperg-Traun and De Boer 1990).

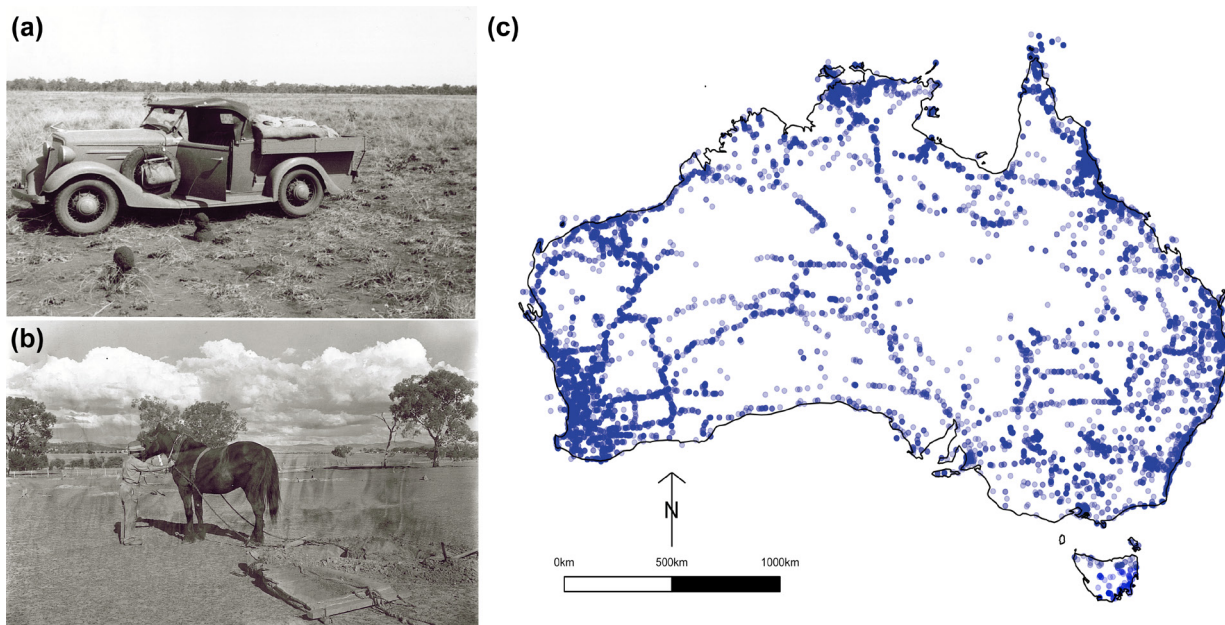


Figure 1. Pictures from termite collection campaigns (a) in cars (1940s) or (b) on horses (1930) excavating an underground termite nest (sourced from the CSIRO). (c) Map of termite species records across Australia (blue point corresponds to a termite species occurrence).

Termite trait data

A literature search was performed on each of the 258 termite species named in the geographic data set. The results of each search were parsed for functional trait information. This resulted in nesting and/or feeding trait data for 182 termite species. The nesting traits include nest type: aboveground mound, belowground, inside tree, arboreal outside tree and in another termite species' nest. Feeding guild categories were based on Donovan et al. (2001) excluding fungus growers which are absent in Australia and soil feeders which are almost non-existent. These data primarily, but not entirely, came from the original taxonomic descriptions of the species as for many species there has been little or no subsequent scientific investigation (Supporting information).

Environmental data

Interpolated grids of mean daily temperature (MDP) and mean annual precipitation (MAP) across Australia were obtained from the Bureau of Meteorology (Bureau of Meteorology 2019). The mean values were calculated from daily temperature and rainfall records between 1961 and 1990 from 6000 stations across Australia. Long-term climate data were extracted for each termite observation using the 'raster' library in R (Hijmans et al. 2021). Temperature and rainfall changed unevenly across the continent during the last century (Hughes 2003), but we assume that the large continental gradients in rainfall and temperature considered are representative of the overall climate across Australia in the past century.

The net primary productivity (NPP) was calculated from MODIS satellite imagery data operated by NASA (Running et al. 2015) with a spatial resolution of 1 km. NPP was calculated on a yearly basis and averaged over 15 years between 2000 and 2015. As for rainfall and temperature, we assume that the large continental gradients we report make uneven local changes in NPP over the past century irrelevant to the patterns we observed. A map of percentage clay in the soil across Australia was obtained from the Soil and Landscape Grid of Australia (Grundy et al. 2015). We used the grid layer of percentage clay in the first 5 cm of the soil; this was highly correlated with deeper layers.

Analysis

To understand the prevalence of termite species sharing certain feeding and nesting traits, the number of species in each feeding and nesting category was counted and the final percentage of species within each group calculated according to the total number of species for which traits were obtained. Some species were documented to be in multiple feeding guilds or nesting types and thus are counted in multiple categories.

Guild-climate relationships

To understand the distribution of termite traits along a climate and soil gradient, generalized linear models (GLM) were used in R. Due to the low correlation between the presence

and absence of most termite traits across the continent (Supporting information), we ran a separate GLM for each trait fitting a binomial distribution with a logit link function. The presence or absence of the trait was used as a response variable against mean annual temperature, precipitation and their interaction, and for nesting traits, a percentage content of clay in the soil was also added. A logarithm (base 10) was applied to the rainfall data for most traits which led to a better fit for all analyses except for inside tree nesting and litter feeding, and for those two traits, untransformed precipitation was used. The fitness of each model was determined from their Akaike information criterion (AIC), the model with the lowest AIC was considered a better fit. Additionally, we checked the homogeneity of the variance and normality for each model by visually assessing plots of the residuals against the fitted values of the model. The models estimated, at a given macroclimate, the probability of a termite occurrence being from a particular guild. Thereafter, to project the distribution of each trait across Australia, the predicted values from the model output for each trait were projected onto the combined temperature and rainfall map of Australia; this is similar to a species distribution model and is sometimes called a trait distribution model (Chen and Smith 2018).

Spatial autocorrelation of each trait was tested by assessing correlation within the residuals of the models using Moran's I through the package 'lctools' in R (Kalogirou and Kalogirou 2015). Moran's I is the most widely used measurement of spatial autocorrelation, and its method is detailed in Sokal and Oden (1978). We are careful to note that the use of a different test for each trait increases the chance of a type-I error and have therefore also run a multivariate model using a redundancy analysis (RDA) to account for some of the collinearity in our response variables (Supporting information). Each sample was considered as a site with a set of traits which were transformed using a 'Hellinger' transformation (Legendre and Gallagher 2001) and modelled against the environmental variables precipitation and temperature.

Species richness and functional diversity

To estimate the species richness and functional diversity of termites in different parts of Australia, the termite distribution data were split into a 2.5 by 2.5 degree grid offering the finest scale grid within which most cells have been sampled for termites (Abensperg-Traun and Steven 1997). The species richness and functional diversity were calculated for each grid. Although the CSIRO expedition effort was considerable, as with all continental scale data sets, heterogeneity in sampling effort was extensive. To account for this heterogeneity, we used rarefaction and extrapolation indexes based on Hill numbers, specifically species richness meaning of order $q = 0$, developed by Chao et al. (2014). This analysis was done using the R package 'iNEXT' (Hsieh et al. 2016). A sample-by-species incidence matrix was built for each grid cell considering each record as a sampling occasion (Supporting information).

Functional diversity was measured using three measures described in Villéger et al. (2008) for each grid cell. Functional richness represents the amount of functional space filled by

the community calculated through the convex hull volume as detailed in Cornwell et al. (2006). Functional evenness describes the evenness of functional traits according to the abundance of each species, and functional divergence represents how abundance is spread along a functional trait axis (Villéger et al. 2008). The abundance of each species within each grid cell was assumed as the number of records for that species; therefore, we assume that the more a species has been recorded, the more abundant it is. All metrics were calculated in R using the 'FD' package (Laliberté et al. 2014). The trait database was combined with the species composition and abundance of each grid cell to compute the different functional diversity measures. The average precipitation, temperature and net primary productivity were calculated for each grid cell. Finally, a correlation matrix using Spearman's rank correlation coefficients, and a correlation test to describe the significance of the correlation with $\alpha=0.05$ as a threshold, was computed between the environmental variables, functional diversity measures and species richness.

From the grid map, we used a hotspot analysis to identify areas with the highest termite species richness to understand the functional diversity of termites across Australia. We characterized species richness hotspots as areas with grid cells that were equal to or over the highest 5% of estimated species richness in accordance with previous studies showing that 1–5% of land area can represent a substantial portion of terrestrial species (Orme et al. 2005). For each species richness hotspot, defined as grid cells over 72 species within 3 grid cells of each other, the average estimated species richness was calculated. Each hotspot was assigned to one of the Whittaker's biomes (Whittaker 1970) according to their average annual rainfall and daily temperature using the 'plotbiome' R package (Valentin and Levin 2018). Finally, the proportion of each feeding and nesting trait was calculated for each hotspot.

Results

Across Australia, the majority of termite species nest inside trees and feed on wood (Table 1). 20.3% of termite species that feed on wood nest inside trees, while 17.6% nest belowground. In total, 51.1% of termite species adopt a belowground nesting strategy, 18% an aboveground mound strategy and 57.1% of termite species feed on wood.

Table 1. Percentage of Australian termite species in the database with particular combinations of feeding and nesting traits.

	Feeding trait			
	Grass feeding (%)	Wood feeding (%)	Litter feeding (%)	Dung feeding (%)
Nesting trait				
Aboveground mound	4.9	5.5	4.4	2.7
Belowground	9.9	17.6	11.0	11.5
Inside trees	0.5	20.3	3.3	7.1
Canopy of trees	0.0	1.1	0.0	0.5
Other termite's nest	2.7	12.6	7.7	7.7

Guild–climate relationships

Due to the high sample size and the relatively low Moran's I after accounting for correlation within the residuals of the trait-based models, we assume that none of the models exhibited spatial autocorrelation (Supporting information). As indicated by the GLM output (Supporting information), the probability of occurrence of wood-feeding species increased with precipitation (Fig. 2) from 0.4 in the subtropical deserts of Northwest Australia (MAP=350 mm) to 1 in Tasmanian temperate rainforests (MAP=2750 mm) (Fig. 3c). The probability of litter and dung-feeding termite species showed the opposite pattern; they increased as precipitation decreased reaching respectively 0.62 in the hot and dry Simpson Desert (MAP=145 mm) (Fig. 3d) and 0.67 in subtropical deserts of Northwest Australia (Fig. 3e). The probability of grass-feeding termites did not change systematically with precipitation although it increased from 0.02 in Tasmanian temperate rainforests (MDP=4°C) to 0.68 in subtropical deserts of West Australia (MDP=26°C) (Fig. 3f) along a rising temperature gradient (Fig. 2).

In line with grass-feeding termite species, the probability of occurrence of belowground nesting species decreased along a temperature and precipitation gradient (Fig. 2) from 0.80 in the hot and dry Strzelecki desert (MAP=130 mm, MDP=21°C) to 0.06 in Tasmanian temperate rainforests (Fig. 3g). Opposite to that observed for belowground feeding species, the probability of finding termites that nest in aboveground mounds was higher at high temperature and high precipitation (Fig. 2), peaking at 0.58 in subtropical deserts of Northwest Australia (Fig. 3h). The probability of aboveground mound nests was higher in lower clay content soils (Fig. 2) in contrary to belowground nests with higher probability of occurring in higher clay content soils (Fig. 2). As aboveground mound nest traits and belowground nest traits exhibited the opposite trends, so too did inside tree versus outside tree nesting traits. As temperatures increase, the probability of finding termite species nesting outside trees increased culminating at 0.31 in the Northeastern rainforest ecosystems (Fig. 3i), while for termite species nesting inside trees, it decreases from 0.96 in Tasmanian temperate rainforests and Southeast Australian temperate seasonal forests (MAP=1700 mm, MDP=4°C) to 0.12 in the subtropical deserts of Northwest Australia (Fig. 3j). The results are also broadly reflected in the RDA biplot (Supporting information).

Species and functional diversity

Species and functional richness did not respond to net primary productivity although functional evenness decreased along that same gradient (Table 2). Additionally, functional evenness and functional richness both increased with temperature (Table 2), while functional evenness decreased along a precipitation gradient. Functional richness and species richness were positively correlated although functional evenness and species richness were negatively correlated (Table 2).

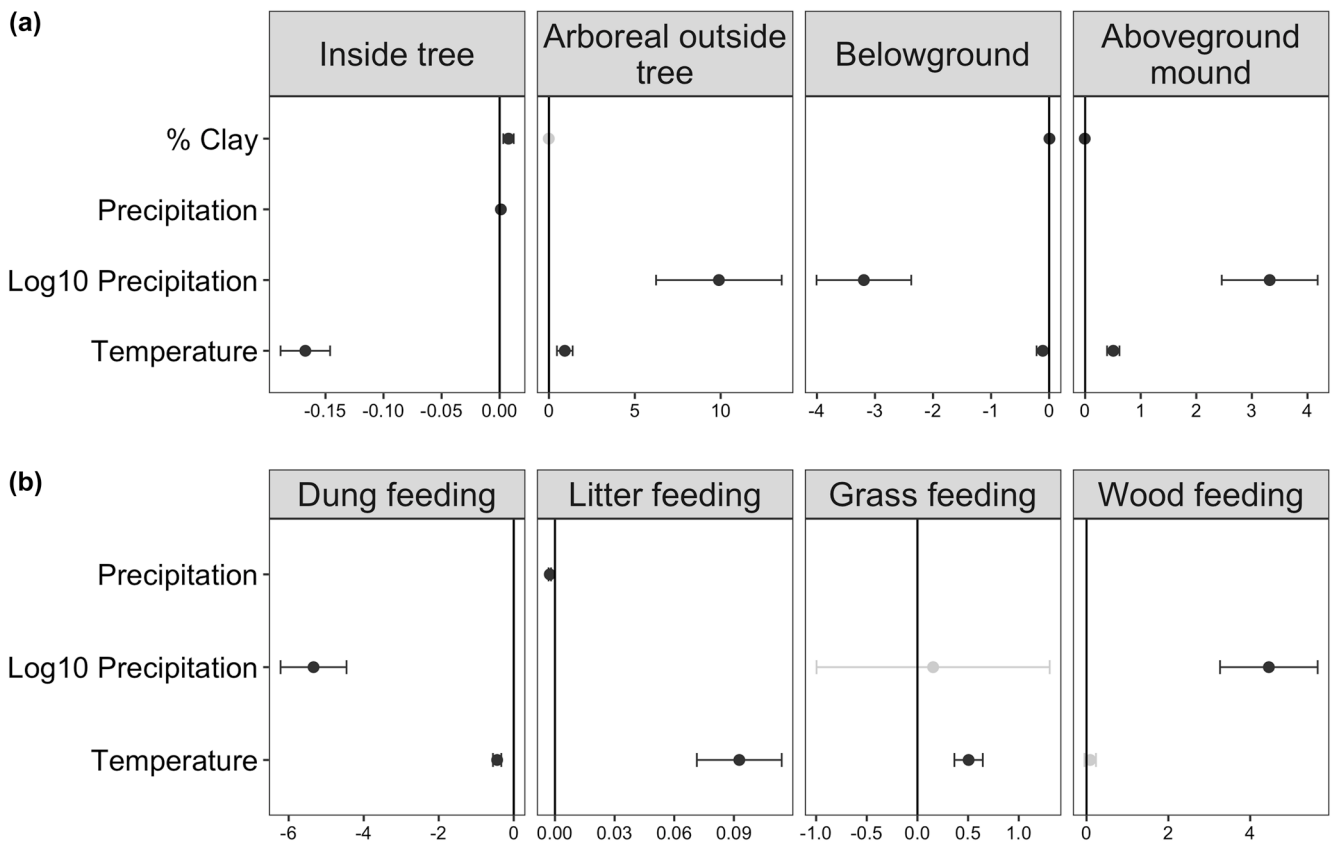


Figure 2. Coefficient ($\pm$ 95% confidence interval (CI)) of termite species (a) nesting traits and (b) feeding traits according to average daily temperature, regular or log (base 10) of average annual precipitation and percentage clay in the soil. No coefficients were calculated for feeding traits against soil properties as there is no basis for it. Light grey circles and black circles represent respectively non-significant and significant coefficients.

The species richness of termites was highest in four areas across Australia: northeast Queensland (91.5 species estimated and 56 observed), north Northern Territory (76 species estimated and 52 observed), west Western Australia (75.4 species estimated and 44.6 observed) and south New South Wales (75.5 species estimated and 28.5 species observed) (Fig. 4a). Four grids contained no termite occurrence records.

Each of the species richness hotspots were found in different climates (Fig. 4b) and have different trait assemblages (Fig. 4c). The species richness hotspot in northeast Queensland (tropical season forests) exhibited the highest average annual precipitation of 1580 mm with 42% of its termite species nesting in aboveground mounds and 5.6% arboreally, on the outside tree. The hotspot in north Northern Territory (tropical savannahs) had the highest average daily temperature with 27.3°C and had the highest proportion of aboveground mound nesting (42.4%) and outside tree nesting species (6.6%). The hotspot in west Western Australia (subtropical deserts) had the lowest average annual precipitation with 342 mm and had the highest proportion of termite species that feed on grass (33%). The hotspot in south New South Wales (ecotone with boundary between temperate grasslands and shrublands) had the lowest average daily temperature with 15.8°C, with 97% of termite species that feed on wood and 79% nesting inside trees.

Discussion

Like better-studied taxa, termites show shifts in guild composition across macro-scale climatic gradients, with important implications for different ecosystems. However, unlike many other taxa, neither the functional nor species richness of termites responds to net primary productivity (Table 2), and these guild frequency shifts are layered onto a complex diversity pattern, with multiple hotspots in different climate zones across Australia. Considered together, the macro-level shifts in termite traits (Fig. 2, 3, Table 1) and species richness imply that each hotspot of species richness has a different mix of dominant termite traits (Fig. 4). The drivers of these shifts may be related to the dominant vegetation type available within different climate zones as well as productivity, temperature and the amount of rainfall.

Unlike many taxa, neither the functional nor species richness of termites responds to net primary productivity (Table 2, Fig. 4a). The lack of a simple relationship with productivity is however largely consistent with predictions by Eggleton et al. (1994) that evolutionary history rather than productivity should drive termite diversity. Termites have diversified throughout the Australian continent from a few colonisation events in the northeast of Australia (Arab et al.

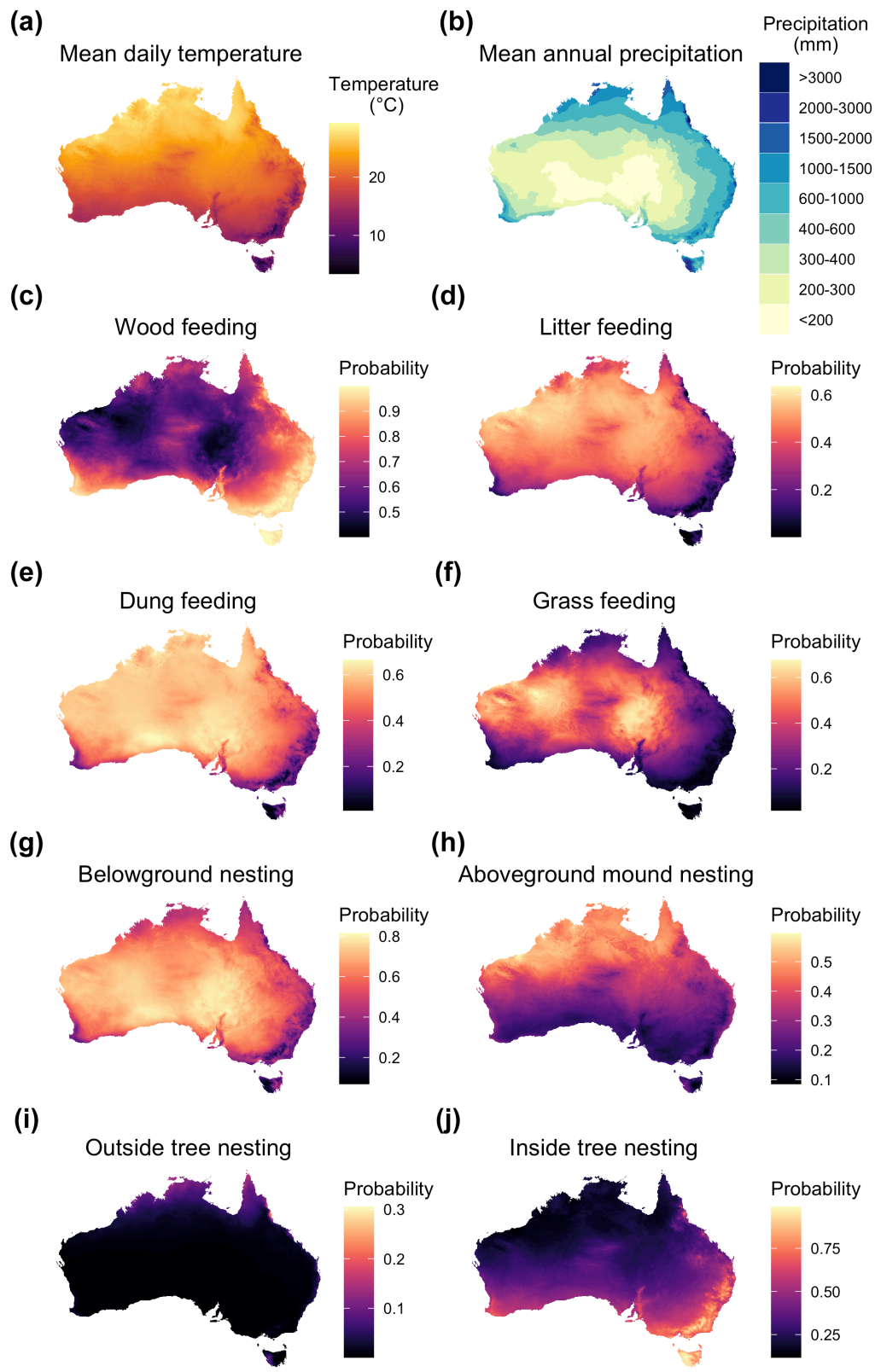


Figure 3. (a) Mean daily temperature (°C) and (b) mean annual precipitation (mm) across Australia. Trait distribution model according to precipitation and temperature for (c) wood feeding, (d) litter feeding, (e) dung feeding, (f) grass feeding and clay content added in for (g) belowground nesting, (h) aboveground nesting, (i) outside tree nesting and (j) inside tree nesting termite species. Note that intraspecific variation is included so a species that eats both litter and dung would be included in both panels, so the proportions at each site may sum to more than 1.

Table 2. Correlation matrix between functional richness, species richness and environmental variables. Each cell corresponds to a correlation coefficient, and asterisks next to the numbers indicate the level of significance of the correlation (* < 0.05, ** < 0.01, *** < 0.001). Red shaded cells indicate significant positive correlations, while blue shaded cells indicate negative significant positive correlations. NPP signifies net primary productivity.

Parameters	Estimated species richness	Observed species richness	Functional evenness	Functional richness	% Clay	NPP	Precipitation	Temperature
Functional divergence	-0.12	-0.14	0.09	0.05	-0.1	-0.34**	-0.21	0.21
Temperature	0.19	0.16	0.31**	0.46***	-0.04	-0.31***	0.12	
Precipitation	0.14	0.25	-0.31*	0.08	0.31**	0.70***		
NPP	0.08	0.21	-0.42***	-0.07	0.12			
% Clay	-0.1	-0.08	0.008	-0.2				
Functional richness	0.70***	0.76***	-0.08					
Functional evenness	-0.42***	-0.47***						
Observed species richness	0.88***							

2017, Lee et al. 2017, Heimbürger et al. 2021). Termites exploit untapped resources in the least productive environments such as grasses low in nutrients (Morton and James 1988) and are highly efficient in buffering against water loss in dry habitats (Dawes 2010, Gautam and Henderson 2014). The combination of these factors has allowed termites to diversify equally in less productive environments compared

to highly productive areas where competition from other established detritivore species could be higher.

Our data also add several important features to the understanding of Australian and global termite diversity. For instance, functional richness is positively correlated with temperature. Interestingly, while functional richness increases with temperature, so does functional evenness which indicates

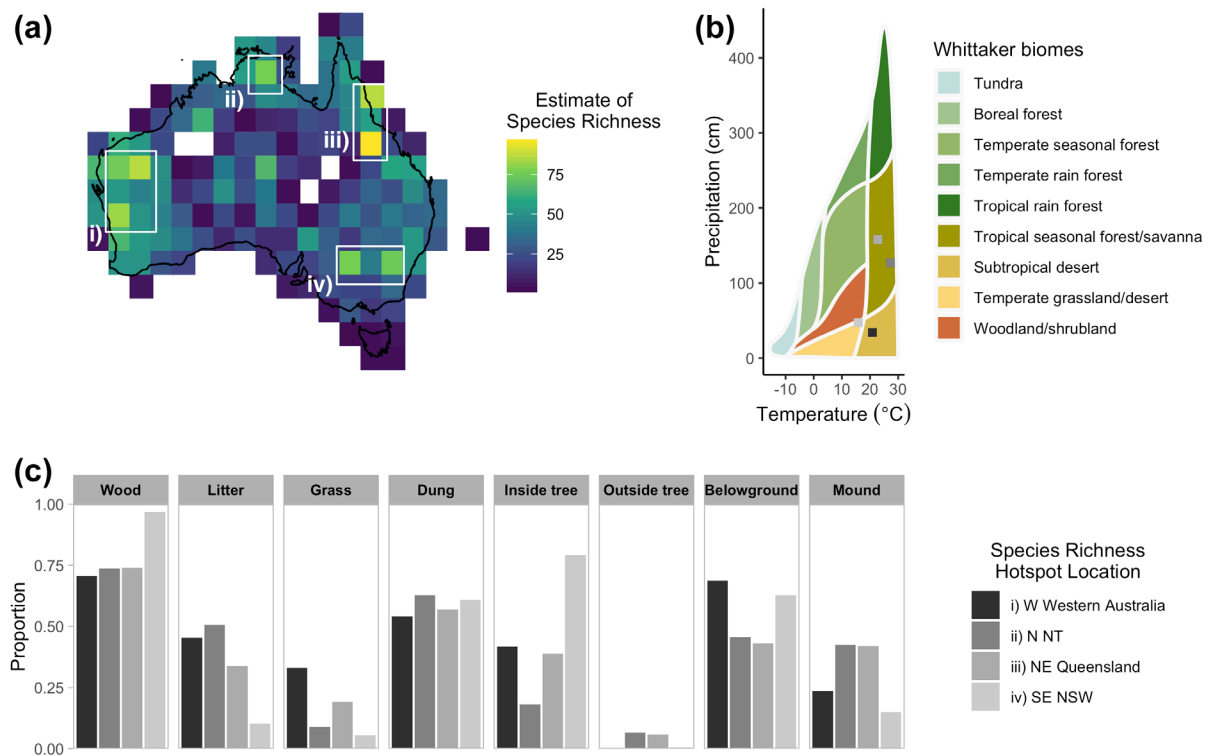


Figure 4. (a) Estimate of termite species richness across Australia on a grid size of 2.5 by 2.5 degrees. Each species richness hotspot area is demarcated with white squares and annotated from i) to iv) corresponding to the names found under species richness hotspot location in the legend. (b) Average climatic conditions for each hotspot of termite species richness (grey squares following the species richness hotspot location legend) overlaid on the Whittaker biomes and the (c) corresponding proportion of feeding and nesting traits for each hotspot which sums up to more than one due to each termites species having several feeding and nesting traits. N, E, W and S correspond respectively to the four cardinal directions north, east, west and south. 'NT' refers to Northern Territory and 'NSW' to New South Wales.

a diversification of functional roles showing the importance of temperature as an environmental filter for termite guild composition such as is seen in ants for instance (Reymond et al. 2013, Bishop et al. 2016). As ectotherms, termites depend on a minimum temperature threshold to function (Sponsler and Appel 1991) while having to buffer against extreme temperatures (Janowiecki et al. 2020); therefore, it is not altogether surprising that temperature acts as an environmental filter on their trait diversification including using different nest structures for instance. Functional evenness decreases with precipitation and net primary productivity suggesting that while the warmest ecosystems have the highest richness and evenness, as it gets wetter and more productive, evenness is lost and more specialized termites proliferate. The loss of functional evenness in wetter ecosystems could be driven by a very high proportion of wood-feeding termites (Fig. 2, 3).

Each termite species richness hotspot has a different trait guild composition following trends in guild composition across macroclimates (Fig. 2c). The only consistent feature of all ecosystems is the presence of wood-feeding termites as the main strategy across all environments. There are two potential explanations for this. First, globally, termites are the only group of animals that can utilize in large amounts the energy stored in wood (Cornwell et al. 2009), and as such, they are effectively only competing with fungi and bacteria for this resource and not other animals while other termite feeding guilds do. Second, biogeographic history may play a role. Wood-feeding species are effective dispersers across water bodies due to the floating properties of logs on which they live (Eggleton and Tayasu 2001, Jones and Eggleton 2011) supporting the hypothesis that the first termite species to arrive on the continent of Australia ate wood and that phylogenetic structure is still strong due to the slow evolution of primitive wood-feeding species (Inward et al. 2007). African and Southeast Asian termite communities have more diverse and equally represented feeding guilds with fungus, hummus and soil feeders as opposed to Australia which is mainly dominated by primitive wood-feeding species providing further evidence to the recent arrival and diversification of termites in Australia (Inward et al. 2007, Jones and Eggleton 2011).

While wood-feeding termites are dominant throughout the Australian continent, termite communities are increasingly composed of wood-feeders where precipitation is high regardless of temperature (Fig. 4a). Wood-feeding termites have their highest probability of occurring along the coast, which spans the range of forest types from wet tropical forests in far north Queensland to the seasonally cold forests in Tasmania (Specht 1981) hypothetically due to a higher abundance of woody resources. Grass, litter and dung-feeders are much less likely to occur in forest ecosystems (Fig. 4b–d). The absence of grass feeders in forests is likely due to the low grass biomass, while the relative lack of litter- and dung-feeders in forests is more intriguing and could be due to higher competition from other invertebrate detritivores (Berg and Bengtsson 2007, Gessner et al. 2010). The dominance of wood-feeding species in wetter environments is therefore also amplified by the lack of other feeding types. On other continents such as

Africa, wetter areas tend to be dominated by soil-feeding termites (Davies et al. 2012) but the low diversity of this clade in Australia seems to favour wood-feeding species.

In contrast, across the arid zone, grass, dung and litter feeders are widespread (Fig. 4b–d). The odds of finding grass-feeding termites increase with temperature while showing no response to rainfall which incidentally coincides with the range of spinifex *Triodia* spp. grasslands which are found across Australia's subtropical deserts (Specht 1981). Litter- and dung-feeding termite species occur at a consistent, low frequency across the Australian continent potentially explained by the widespread availability of these food sources. Dung- and litter-feeding strategies go hand in hand within termite species as both sources are nutritionally low but combined they offer beneficial opportunistic feeding methods (Freyman et al. 2008). In all, the distribution of termite-feeding guilds throughout the Australian continent reflects the distribution of available resources. While previous evidence supports the fact that wood-feeding termites tend to respond to wood availability (Maynard et al. 2015), our study provides further evidence that termites belonging to other feeding guilds diversify along macroclimate gradients shaped by resource availability.

The probability of finding belowground nesting species is higher in the drier desert environments (Fig. 4e) of Australia similar to the pattern for litter feeding and dung-feeding termite species. Aboveground mound nesting species, on the other hand, are more readily found in ecosystems with higher precipitation and temperature coinciding with savannah woodlands (Specht 1981) (Fig. 4f). Interestingly, previous studies on the evolution of mound building in *Coptotermes* (Lee et al. 2017) and *Nasutitermes* (Beasley-Hall et al. 2018) in Australia suggested that macroclimate is a negligible force in the distribution of aboveground nesting termite species. Instead, the studies speculate that biological factors such as predator–prey relationships or the evolution of new foraging behaviours are more important drivers of mound evolution. While the previous studies on the evolution of mound building in Australia only focused on a specific genus, our study combined all the genera producing a clear pattern on the climatic influence on the trait.

Termite individuals are highly susceptible to water loss (Collins 1969) making thermoregulation and humidity control a crucial factor determining the design of their nests explaining why macroclimate may limit the establishment of aboveground mound nests. Aboveground mound nests require complex ventilation mechanisms to maintain homeostatic conditions within the nest (Noirot and Darlington 2000, Korb 2003) leading to high evapotranspiration rates (Korb 2011). Low precipitation leading to low water availability in dry desert landscapes therefore prevents the development of aboveground mound nests. Additionally, low nocturnal winter temperatures in southern dry desert landscapes as opposed to their significantly higher temperatures in tropical areas, emphasize the need for termites to insulate their nests to avoid excessive temperature loss. The more stable environment found in the soil therefore favours belowground

nesting in southern dry desert landscapes (Korb 2011). The macro-climatic limits to aboveground protruding nest establishment are also reflected in the distribution of outside tree nests which are disproportionately found in tropical rainforests (high precipitation and high temperature).

The variability in the distribution of nesting strategies has important trophic implications as it has repercussions for the distribution of other species. Different nest types are accessible to various functional predator groups therefore playing distinct roles in trophic webs. The higher soil to nest interface in belowground nests favours access by subterranean predators such as ants or legless lizards *Lerista* spp., while aboveground mound nests are more visible and accessible to large mammalian termite-feeding specialists such as short-beaked echidnas *Tachyglossus aculeatus* or numbats *Myrmecobius fasciatus*. Many organisms around the world, of which a few are under threat such as giant anteaters *Myrmecophaga tridactyla* in South America or the giant pangolin *Smutsia gigantea* in Africa, depend on termites as a food source (Redford 1987) therefore making it of utmost importance to understand the macroecology of termites to aid in their conservation.

Understanding the biogeography and macroecology of termites is just beginning. The lack of studies seems largely due to their cryptic nature, since they are clearly a crucial group for ecosystem function. For instance, the high frequency of belowground nesting termite species in southeast Australia combined (Fig. 3g) with the low sampling effort can explain the discrepancy in the observed species richness in the area by Abensperg-Traun and Steven (1997). Although we used a modern technique to account for sampling bias, the species richness hotspots still partially coincide closely with areas exhibiting higher sampling efforts and easier access by CSIRO scientists, due to their proximity to old field stations, therefore emphasizing the need to interpret these results with caution. The biased nature of termite sampling in Australia has led to the underestimation of termite species in large areas of Australia (Supporting information); therefore, we advocate for higher sampling to be undertaken in regions such as southeast Australia and Western Australia.

All in all, our study highlights the unconventional patterns in species and functional richness of termites which do not increase with primary productivity while also highlighting the importance of temperature acting as an environmental filter in their diversification. Additionally, we described the shifts in termites' traits across macroclimate gradients. Patterns in feeding guild distributions are shaped by resource availability, while mound-building termites prefer warmer and wetter climates for their establishment. While our study pertains to Australian ecosystems, the current lack of studies on the macro-ecology of termites on other continents such as South America or Africa warrants for further investigations of termite trait responses to macroclimate gradients around the globe to test the generality of our findings. In the current changing climate (IPCC 2014), it is crucial to predict potential changes in ecosystem functions of which termites are one of the most abundant groups alongside ants, especially in the tropics where they represent up to 10% of all animal biomass

(Lavelle et al. 1997) and contribute to over 50% of all wood decomposition (Griffiths et al. 2019). Our research opens avenues to explore the current and potential distribution and functional role of termites on regional and global scales having far-reaching implications for the distribution of many other species which depend on termites.

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Author contributions

Baptiste J. Wijas: Conceptualization (equal); Data curation (equal); Formal analysis (lead); Investigation (equal); Methodology (lead); Visualization (lead); Writing – original draft (lead). **Shevaun Lim:** Data curation (equal); Resources (lead). **William K. Cornwell:** Conceptualization (equal); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Supervision (lead); Writing – review and editing (lead).

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Data availability statement

Data are available from the Dryad Digital Repository: <<https://doi.org/10.5061/dryad.2jm63xsqk>> (Wijas et al. 2021).

Supporting information

The supporting information associated with this article is available from the online version.

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